

URITECT Sex-Specific Bayesian UTI Parameter Review Form v1.3

Female model: uti_bayesian_female_v1_1_20260926 **Male candidate:** uti_bayesian_male_v0_1_20261001 **Purpose:** Expert content review, not clinical validation

Reviewer Brief

URITECT is an offline smartphone urinalysis screening and decision-support application. It uses separate literature-derived female and male research models. Please review whether the proposed laboratory mappings, evidence parameters, and limitations are relevant and clearly implemented.

The output is not a diagnosis, treatment recommendation, or replacement for urine culture or professional judgment. Further clinical and statistical validation is required before the percentages can be interpreted as clinically validated estimates.

Female Parameters

ID	Finding or pattern	Candidate value	Source
F01	Starting prior in eligible symptomatic women	50%	Bent 2002; Kurotschka 2024
F02	Nitrite positive	LR 5.50	Kurotschka 2024
F03	Leukocyte esterase positive	LR 1.40	Kurotschka 2024
F04	Dipstick blood positive	LR 1.70	Kurotschka 2024
F05	Nitrite plus leukocytes or blood	LR 7.20	Little 2006
F06	Nitrite, leukocytes, and blood all negative	LR 0.22	Little 2006
F07	Dysuria and urgency	LR 1.50	Kurotschka 2024
F08	Dysuria	LR 1.30	Kurotschka 2024
F09	Urgency	LR 1.20	Kurotschka 2024
F10	Frequency	LR 1.10	Kurotschka 2024

Male Candidate Parameters

ID	Finding or pattern	Candidate value	Source
M01	Younger male subgroup starting prior	52.2%	Derived from den Heijer 2012 Table 3
M02	Nitrite and leukocyte esterase positive	LR 5.14	den Heijer 2012 Table 2
M03	Nitrite-positive threshold	LR 4.87	den Heijer 2012 Table 2
M04	Leukocyte-est-positive threshold	LR 1.65	den Heijer 2012 Table 2
M05	Nitrite and leukocyte esterase both negative	LR 0.35	den Heijer 2012 Table 2

The male values are applied in an ordered hierarchy, with one LR only. The source reports overlapping score thresholds rather than exact mutually exclusive pattern LRs. M03 and M04 are therefore a documented URITECT research approximation requiring statistical and culture-confirmed validation.

Mapping and Use Statements

Rate each statement: 4 = highly relevant/acceptable, 3 = minor revision, 2 = major revision, 1 = not acceptable.

ID	Review statement	1	2	3	4	Comment
C01	Female nitrite, LE, and blood mappings match the stated source thresholds.	☐	☐	☐	☐	
C02	Female Trace 15 LE is neutral because the female source threshold is + or greater.	☐	☐	☐	☐	
C03	Male Trace 15 LE is positive because the male study counted any color change.	☐	☐	☐	☐	
C04	Nitrite requires the strip category Positive in both models.	☐	☐	☐	☐	
C05	Blood is appropriately omitted from the male numerical model.	☐	☐	☐	☐	
C06	One mutually exclusive factor prevents duplicate multiplication of correlated dipstick evidence.	☐	☐	☐	☐	
C07	Missing and unreliable results correctly receive no update.	☐	☐	☐	☐	
C08	The female and male eligibility restrictions are understandable.	☐	☐	☐	☐	
C09	Safety and alternate-cause findings are appropriately handled outside the equation.	☐	☐	☐	☐	
C10	The lack of local calibration and culture-confirmed validation is stated clearly.	☐	☐	☐	☐	

Parameter Ratings

Please separately rate each parameter listed as F01-F10 and M01-M05.

Parameter ID	1	2	3	4	Recommended revision or comment
F01	☐	☐	☐	☐	
F02	☐	☐	☐	☐	
F03	☐	☐	☐	☐	
F04	☐	☐	☐	☐	
F05	☐	☐	☐	☐	
F06	☐	☐	☐	☐	
F07	☐	☐	☐	☐	
F08	☐	☐	☐	☐	
F09	☐	☐	☐	☐	
F10	☐	☐	☐	☐	
M01	☐	☐	☐	☐	
M02	☐	☐	☐	☐	
M03	☐	☐	☐	☐	
M04	☐	☐	☐	☐	
M05	☐	☐	☐	☐	

Overall Decision

☐ Acceptable as provisional literature-derived research parameters ☐ Acceptable with minor revisions ☐ Major revision required before research use ☐ Unable to assess

Comments: _____

Acknowledgment

I reviewed the proposed literature-derived parameters, strip mappings, and screening logic for their stated research purpose. I understand that this expert content review does not independently establish diagnostic accuracy, clinical calibration, sensitivity, specificity, or clinical validity of the complete URITECT models.

Reviewer name: _____ Professional position: _____

PRC license number: _____
Institution: _____
Signature: _____ Date: _____

References

1. den Heijer CDJ, et al. *Br J Gen Pract*. 2012;62:e780-e786. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3481519/>
2. Koeijers JJ, et al. *Clin Infect Dis*. 2007;45:894-896. <https://pubmed.ncbi.nlm.nih.gov/17806056/>
3. Kurotschka PK, et al. *Am Fam Physician*. 2024;109:167-174.
4. Little P, et al. *Br J Gen Pract*. 2006;56:606-612.
5. Bent S, et al. *JAMA*. 2002;287:2701-2710.